

# 17 - Cascade vs Static: Cumulative-FLOPs Comparison

This document compares every cascade we ran against the **single static (fixed-budget) models**, under correct **cumulative** FLOPs accounting (an image that exits at a stage has already run all earlier stages). "Exit-only" FLOPs (charging only the exit model) are shown too, but cumulative is the true cost.

"**Beats static**" means a cascade point gives **higher (or equal) accuracy at lower FLOPs** than any single fixed-budget model. All FLOPs are GFLOPs; % is relative to the dense model of that dataset.

## A. CIFAR-100 - cascade 25 -> 50 -> 75 -> dense

Static models (reference):

| Model | Accuracy | FLOPs | % dense |
|-------|----------|-------|---------|
| 25%   | 75.83%   | 0.687 | 64%     |
| 50%   | 78.18%   | 0.818 | 76%     |
| 75%   | 79.16%   | 0.949 | 88%     |
| dense | 79.73%   | 1.079 | 100%    |

Best cascade operating points (thresholds = [t25, t50, t75]):

| What                         | Thresholds      | Accuracy      | Exit-only (% dense) | Cumulative (% dense) |
|------------------------------|-----------------|---------------|---------------------|----------------------|
| Best accuracy                | [0.9, 0.9, 0.9] | <b>81.82%</b> | 0.763 (71%)         | 1.208 (112%)         |
| Highest acc <= dense compute | [0.9, 0.8, 0.4] | 81.37%        | 0.747 (69%)         | 1.078 (100%)         |
| Cheapest matching dense acc  | [0.7, 0.6, 0.4] | 79.73%        | 0.717 (66%)         | <b>0.882 (82%)</b>   |
| Most efficient               | [0.3, 0.3, 0.3] | 76.29%        | 0.689 (64%)         | 0.699 (65%)          |

**This cascade beats static.** At [0.7,0.6,0.4] it matches dense accuracy (79.73%) at only **82% of dense compute** - better than both the dense and 75% static models (110 such combos exist). The ensemble effect even lifts accuracy to **81.37% at equal dense compute**. Exit-only makes it look cheaper still (66%), but 82% cumulative is the honest figure.

**Can it beat each static model?** (a cascade combo with accuracy >= the model **and** lower cumulative FLOPs than it)

| Static model         | Beaten? | Best cascade combo (cumulative)             | Compute saved |
|----------------------|---------|---------------------------------------------|---------------|
| 25% (75.83% @ 0.687) | no      | - (cascade always runs the 25% stage first) | -             |
| 50% (78.18% @ 0.818) | yes     | [0.6, 0.3, 0.3] -> 78.55% @ 0.798 G         | 2.4%          |

|                        |     |                                     |       |
|------------------------|-----|-------------------------------------|-------|
| 75% (79.16% @ 0.949)   | yes | [0.7, 0.3, 0.3] -> 79.27% @ 0.842 G | 11.2% |
| dense (79.73% @ 1.079) | yes | [0.7, 0.6, 0.4] -> 79.73% @ 0.882 G | 18.2% |

It beats **every static model except the cheapest 25%** - the cascade can never undercut the first stage it always runs. Against 50% / 75% / dense it gives equal-or-higher accuracy at lower cumulative compute (up to 18% saving vs dense).

## B. CIFAR-100 - sub-dense cascade 10 -> 25 -> 50

Static models (reference):

| Model | Accuracy | FLOPs | % dense |
|-------|----------|-------|---------|
| 10%   | 70.82%   | 0.607 | 56%     |
| 25%   | 75.83%   | 0.687 | 64%     |
| 50%   | 78.18%   | 0.818 | 76%     |

Best cascade operating points (thresholds = [t10, t25]):

| What                        | Thresholds   | Accuracy | Exit-only (% dense) | Cumulative (% dense) |
|-----------------------------|--------------|----------|---------------------|----------------------|
| Best accuracy               | [0.98, 0.90] | 79.57%   | 0.680 (63%)         | 1.155 (107%)         |
| Highest acc < dense compute | [0.98, 0.70] | 79.11%   | 0.666 (62%)         | 1.066 (99%)          |
| Most efficient              | [0.30, 0.30] | 71.26%   | 0.609 (56%)         | 0.616 (57%)          |

**This cascade does NOT beat static.** To reach ~79% it needs ~99% of dense compute, but the single 75% model already gives 79.16% at 88%, and the 50% model gives 78.18% at 76%. At every accuracy level a single fixed budget is cheaper - static is the frontier.

Can it beat each static model?

| Static model         | Beaten? |
|----------------------|---------|
| 10% (70.82% @ 0.607) | no      |
| 25% (75.83% @ 0.687) | no      |
| 50% (78.18% @ 0.818) | no      |

No combo Pareto-beats any static model - the static models are the full efficient frontier.

## C. ImageNet-1K - cascade 25 -> 50 -> 75 -> dense, prune layer 6

Static models (reference, zero-shot DeiT-Small):

| Model | Accuracy | FLOPs | % dense |
|-------|----------|-------|---------|
| 25%   | 71.30%   | 2.686 | 63%     |
| 50%   | 77.74%   | 3.208 | 75%     |
| 75%   | 79.29%   | 3.729 | 88%     |
| dense | 79.71%   | 4.251 | 100%    |

**Best cascade operating points** (thresholds = [t25, t50, t75]):

| What                         | Thresholds      | Accuracy | Exit-only (% dense) | Cumulative (% dense) |
|------------------------------|-----------------|----------|---------------------|----------------------|
| Best accuracy                | [0.9, 0.9, 0.8] | 79.71%   | 3.969 (93%)         | 11.611 (273%)        |
| Highest acc <= dense compute | [0.5, 0.3, 0.4] | 77.50%   | 2.922 (69%)         | 4.238 (100%)         |
| Most efficient               | [0.3, 0.3, 0.3] | 75.05%   | 2.806 (66%)         | 3.486 (82%)          |

**No cascade combo beats static.** Matching dense accuracy costs **273% of dense** under cumulative accounting. Exit-only (93%) is hugely misleading here - half the images run all four models. Even the efficient point (75.05% @ 82%) is beaten by the static 50% model (77.74% @ 75%).

**Can it beat each static model?**

| Static model           | Beaten? |
|------------------------|---------|
| 25% (71.30% @ 2.686)   | no      |
| 50% (77.74% @ 3.208)   | no      |
| 75% (79.29% @ 3.729)   | no      |
| dense (79.71% @ 4.251) | no      |

No combo beats any static model - cumulative cost rules it out at every level.

## D. ImageNet-1K - cascade 25 -> 50 -> 75 -> dense, prune layer 3

**Static models (reference, zero-shot, layer-3 pruning):**

| Model | Accuracy | FLOPs | % dense |
|-------|----------|-------|---------|
| 25%   | 67.57%   | 1.904 | 45%     |
| 50%   | 76.99%   | 2.686 | 63%     |
| 75%   | 79.12%   | 3.469 | 82%     |
| dense | 79.71%   | 4.251 | 100%    |

**Best cascade operating points** (thresholds = [t25, t50, t75]):

| What                         | Thresholds      | Accuracy | Exit-only (% dense) | Cumulative (% dense) |
|------------------------------|-----------------|----------|---------------------|----------------------|
| Best accuracy                | [0.9, 0.9, 0.8] | 79.71%   | 3.824 (90%)         | 10.046 (236%)        |
| Highest acc <= dense compute | [0.7, 0.4, 0.3] | 78.27%   | 2.501 (59%)         | 4.154 (98%)          |
| Most efficient               | [0.3, 0.3, 0.3] | 73.08%   | 2.107 (50%)         | 2.696 (63%)          |

**Still no cascade combo beats static.** Pruning earlier makes each stage cheaper (the curve shifts down vs layer 6), but the best sub-dense point (78.27% @ 98%) is beaten by the static 75% model (79.12% @ 82%), and matching dense accuracy still costs 236%.

#### Can it beat each static model?

| Static model           | Beaten? |
|------------------------|---------|
| 25% (67.57% @ 1.904)   | no      |
| 50% (76.99% @ 2.686)   | no      |
| 75% (79.12% @ 3.469)   | no      |
| dense (79.71% @ 4.251) | no      |

Earlier pruning lowers each stage's cost but still no combo Pareto-beats any static model.

## Overall summary

| Cascade                       | Beats static (cumulative)? | Best result                                                               |
|-------------------------------|----------------------------|---------------------------------------------------------------------------|
| CIFAR 25->50->75->dense       | Yes                        | 79.73% (= dense) at <b>82%</b> of dense; ensemble lifts to 81.37% at 100% |
| CIFAR 10->25->50 (sub-dense)  | No                         | best 79.11% at 99%; static 75% gives 79.16% at 88%                        |
| ImageNet L6 25->50->75->dense | No                         | matching dense costs 273%; static always cheaper                          |
| ImageNet L3 25->50->75->dense | No                         | best 78.27% at 98%; static 75% gives 79.12% at 82%                        |

**Take-away:** the cascade only beats static in **one** case - CIFAR-100 with the full 25->50->75->**dense** ladder - and it wins there because of the **ensemble lift** (routing across four differently-trained models, including dense, raises accuracy above any single model). Whenever the strong/dense stages are removed (sub-dense) or the data is hard and zero-shot (ImageNet), the cumulative cost dominates and a single static budget is the efficient frontier.